INCH-POUND

MIL-M-14/8
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, EPOXY THERMOSETTING, CONTAINING MINERAL/GLASS FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Epoxy resin molding compounds, mineral/glass filled, shall be of the following types, as specified:

Type MEC	–	Mineral filler, encapsulation grade, for molding at extremely low pressure
Type MEE	–	Mineral filler, low pressure molding, electrical grade
Type MEG	–	Mineral filler, general purpose
Type MEH	–	Mineral filler, heat resistant
Type GEI-5	–	Glass fiber filler, 0.5 foot-pound per inch notch
Type GEI-20	–	Glass fiber filler, 2.0 foot-pounds per inch notch
Type GEI-100	–	Glass fiber filler, 10.0 foot-pounds per inch notch
Type GEI-250	–	Glass fiber filler, 25.0 foot-pounds per inch notch.

REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

AMSC N/A FSC 9330

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NOTES

Type MEC. This type is a mineral-filled epoxy compound intended for encapsulation purposes.

Type MEE. This type is a mineral-filled epoxy compound for use in encapsulation with good dielectric properties and arc and flame resistance.

Type MEG. This type is a mineral-filled, general purpose, epoxy compound.

Type MEH. This type is a mineral-filled, epoxy compound intended for applications requiring heat resistance.

Type GEI-5. This type is a glass-filled epoxy compound of moderate-impact strength.

Type GEI-20. This type is a glass-filled epoxy compound with good impact strength.

Type GEI-100. This type is a glass-filled epoxy compound having high-impact strength.

Type GEI-250. This type is a glass-filled epoxy compound with very-high-impact strength.

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TABLE I. *Qualification test requirements for epoxy resin molding compounds, mineral/glass filled.*

Requirement	Type MEC	Type MEE	Type MEG	Type MEH	Type GEI-5	Type GEI-20	Type GEI-100	Type GEI-250
Mechanical/physical								
Compressive strength, endwise	20,000	20,000	20,000	20,000	25,000	25,000	25,000	26,000
Dimensional stability	–	0.1	0.1	0.1	0.1	0.1	0.2	0.2
Flexural strength	12,000	12,000	12,000	12,000	15,000	16,000	16,000	35,000
Heat deflection temperature	130	130	130	175	130	150	175	200
Heat resistance	150	150	150	150	150	150	150	150
Impact strength, side[1]	0.2	0.3	0.3	0.35	0.5	2.0	10.0	25.0
Tensile strength	6,000	6,000	6,000	6,000	7,000	8,000	8,000	15,000
Water absorption	0.5	0.3	0.3	0.3	0.3	0.5	0.5	0.25
Electrical								
Arc resistance	120	180	150	150	150	130	130	120
Dielectric breakdown:								
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)								
Step-by-step test	40	40	40	40	40	40	40	40
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)								
Step-by-step test	35	35	35	35	35	35	35	35
Dielectric constant:								
at 1 kilohertz	5.2	5.0	5.3	5.8	5.8	5.8	5.8	6.3
	5.4	5.2	5.5	6.0	6.0	6.0	6.0	6.6
at 1 megahertz	5.0	4.8	5.5	5.5	5.5	5.5	5.5	6.0
	5.2	5.0	5.7	5.7	5.7	5.7	5.7	6.3

See footnote at end of table.

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TABLE I. *Qualification test requirements for epoxy resin molding compounds, mineral/glass filled – Continued.*

Requirement	Type MEC	Type MEE	Type MEG	Type MEH	Type GEI-5	Type GEI-20	Type GEI-100	Type GEI-250
Electrical *(Continued)*								
Dielectric strength:								
Short-time test	325	325	325	325	325	300	300	300
Step-by-step test	300	300	300	300	300	275	275	275
Short-time test	325	325	325	325	325	300	300	300
Step-by-step test	300	300	300	300	300	275	275	275
Dissipation factor:								
at 1 kilohertz	0.018	0.018	0.020	0.025	0.025	0.025	0.025	0.025
	0.020	0.020	0.025	0.030	0.030	0.030	0.030	0.030
at 1 megahertz	0.018	0.018	0.025	0.020	0.020	0.020	0.020	0.020
	0.020	0.020	0.030	0.025	0.025	0.025	0.025	0.025
Surface resistance	1.0	1,000	6.0	100	1,000	1,000	1,000	1,000
Volume resistance	1.0	1,000	5.0	100	1,000	1,000	1,000	1,000
Combustion								
Flame resistance:								
Ignition time	90	90	90	90	90	90	90	90
Burning time	90	90	90	90	90	90	90	250
Flammability:	V-1	V-1	V-1	V-1	V-1	V-1	V-1	V-1

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for epoxy molding compounds, mineral/glass filled.*

Property to tested	Type MEC	Type MEE	Type MEG	Type MEH	Type GEI-5	Type GEI-20	Type GEI-100	Type GEI-250
Arc resistance	120	180	150	150	150	130	130	120
Dielectric constant at 1 megahertz	5.2	5.0	5.7	5.7	5.7	5.7	5.7	6.3
Dielectric strength, step-by-step	300	300	300	300	300	275	275	275
Dissipation factor at 1 megahertz	0.020	0.020	0.030	0.025	0.025	0.025	0.025	0.025
Flexural strength	12,000	12,000	12,000	12,000	15,000	15,000	16,000	35,000
Impact strength, side[1]	0.2	0.3	0.3	0.35	0.5	2.0	10.0	25.0
Water absorption	0.5	0.3	0.3	0.3	0.3	0.5	0.5	0.25

[1]The side of the test specimen is that area formed by the chase of the mold.

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